

Environmental Provisions in Chinese Preferential Trade Agreements

Do They Green China's Exports?

Working Paper — Project Status Report

DRAFT — For Internal Use Only

Edoardo Vitella

PhD Candidate

University of Trento & Free University of Bozen-Bolzano

June 2026

This document provides a comprehensive description of the project's research question, identification strategy, data, and current empirical status. It is intended as an orientation guide both for the author and for collaborators approaching the project for the first time. All analytical results reported here are preliminary and subject to revision.

Abstract

This paper investigates whether environmental provisions (EPs) in China's preferential trade agreements (PTAs) signed between 2000 and 2015 affect the composition of Chinese exports at the firm level. Using a transaction-level customs dataset covering approximately 49 million firm-product-destination-year observations, combined with two independent indices of environmental provision depth — the World Bank Deep Trade Agreements (DTA) database and the TREND dataset (Morin et al. 2018) — we estimate the causal effect of EP depth on firms' exports of green and pollution-intensive goods.

Our identification strategy exploits within-firm, within-destination, within-year variation across products of different environmental intensity, using a triple-difference design with a firm-destination-year fixed effect (fdt) that absorbs the entire entry-into-force of the agreement — tariff cuts, investment flows, trade facilitation — as a nuisance. Identification comes from the differential response of green versus dirty versus neutral products within the same firm-market-year cell, comparing markets where agreements have deeper environmental provisions.

Preliminary diagnostic results show that the level effect of EP depth on aggregate exports is a precisely-estimated null — a pattern consistent with selection rather than a causal effect. The coefficient attenuates monotonically across progressively more saturated fixed-effect structures, and the significance found in the least-saturated specification is an artefact of under-clustering. The paper's key contribution is to test whether EPs reshape the composition of Chinese exports (green goods up, dirty goods down), using micro-level identification unavailable to prior country-level studies (Brandt et al. 2020; Abman, Lundberg & Ruta 2024).

Keywords: environmental provisions, preferential trade agreements, China, green exports, dirty goods, firm-level customs data, triple-difference, pollution haven hypothesis, EP depth, TREND database, World Bank DTA

JEL Codes: F13, F18, Q56, Q58

1. Introduction and Research Question

Preferential trade agreements (PTAs) have increasingly included environmental provisions (EPs) — ranging from general commitments to maintain domestic environmental laws to specific chapters on biodiversity, climate change, fisheries, and green market access. As of 2016, over 680 PTAs contained at least one environmental provision, and the average number of such provisions per agreement has grown substantially since the early 2000s (Morin, Dür & Lechner 2018). This proliferation raises a fundamental empirical question: are these provisions substantively consequential for trade flows, or are they largely declaratory?

The question is particularly salient for China. As the world's largest goods exporter and a major source of embodied carbon in international trade, China's trade agreements have important environmental implications that extend well beyond its bilateral partners. Between 2000 and 2015, China concluded 14 PTAs with 25 partner countries, several of which contain explicit environmental chapters — though their content and enforceability vary widely. Whether these provisions have any measurable effect on the environmental profile of Chinese exports is an open empirical question that this paper addresses for the first time using firm-level data.

The existing empirical literature on EPs and trade flows is thin and relies entirely on aggregate country-level data. Brandi et al. (2020, *World Development*) — the most direct predecessor — find that EPs in PTAs reduce the share of dirty exports and increase the share of green exports from developing countries, using a country-pair-year gravity panel spanning 1984–2016. Abman, Lundberg & Ruta (2024, *JEEA*) find that regional trade agreements with forest and biodiversity provisions effectively curb deforestation, using satellite data on land-use change. Both studies work at a level of aggregation that prevents them from disentangling the effect of EPs from the broader effects of trade liberalization, and neither can test for within-firm reallocation — arguably the most economically interesting margin.

1.1 Core Research Questions

- Do environmental provisions in China's PTAs shift the *composition* of Chinese exports towards greener goods and away from pollution-intensive goods? (The triple-difference question: the paper's central contribution.)
- Is the aggregate (level) effect of EP depth on Chinese exports a precisely-estimated null — and if so, does this reflect selection of higher-EP partners rather than a genuine causal effect? (The diagnostic/methodological contribution against Brandi 2020 and ALR 2024.)
- What is the role of within-firm reallocation: do multi-product firms shift their export mix towards green goods when serving EP-deep destinations, conditional on the overall level of their bilateral trade relationship?
- Which types of EP — green market access provisions, enforcement mechanisms, hard versus soft obligations — drive the effect, if any?

The paper's comparative advantage is the use of **firm-level Chinese customs data** covering approximately 49 million transactions at the firm-product-destination-year level. This granularity enables an identification strategy unavailable to prior work. The firm-destination-year fixed effect (fdt) absorbs everything that happens when an agreement enters into force — tariff cuts, investment liberalization, trade facilitation provisions, exchange rate effects — leaving only the differential effect of EP depth on green versus dirty versus neutral products within the same firm-market-year cell.

This is the Rajan-Zingales (1998) logic applied to environmental provisions.

The answers carry implications both for the design of trade agreements and for the pollution-haven hypothesis applied to China. A null result — if precisely identified — would suggest that EPs in Chinese PTAs are largely declaratory rather than binding constraints on trade composition, an important finding for international environmental governance. A positive result would provide the first firm-level evidence that environmental treaty commitments reshape the micro-level structure of trade, beyond what aggregate studies can establish.

2. China's PTA Landscape and Environmental Provisions

China's engagement with preferential trade agreements accelerated significantly after its WTO accession in 2001. Prior to that, China had no PTAs in force. By 2015, it had concluded 14 agreements covering partners in Southeast Asia, South Asia, the Middle East, and Oceania. These agreements exhibit substantial heterogeneity in depth, scope, and environmental content.

2.1 China's Agreements in the Sample (2000–2015)

Agreement	Entry in Force	Partner Countries	WB EP Depth (approx.)	Notes
ASEAN-China FTA (ACFTA)	2005–2010	Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, Philippines, Singapore, Thailand, Vietnam	Low–Moderate	Phased implementation; ASEAN as single agreement with 10 partners
China–Pakistan FTA	2007	Pakistan	Low	Goods agreement; limited environmental chapter
China–Chile FTA	2006	Chile	Moderate	Includes environment chapter; supplemented by 2012 upgrade
China–New Zealand FTA	2008	New Zealand	High	Relatively comprehensive; NZ pushed for environmental text
China–Singapore FTA	2009	Singapore	Moderate	Bilateral; overlaps with ACFTA
China–Peru FTA	2010	Peru	Moderate	Includes environment chapter; Latin America
China–Costa Rica FTA	2011	Costa Rica	Moderate	Smaller agreement; EP provisions unclear
China–Iceland FTA	2014	Iceland	Moderate–High	Nordic partner; EU-influenced environmental text

Agreement	Entry in Force	Partner Countries	WB EP Depth (approx.)	Notes
China–Switzerland FTA	2014	Switzerland	High	Comprehensive; includes specific EP obligations
CEPA (HK & Macao)	2003–2004	Hong Kong, Macao	—	Sui generis; entrepôt hubs; excluded from main sample

Note: ASEAN counts as a single agreement for identification purposes, yielding approximately **14 effective treatment clusters** (not 25 partner countries). This is important for inference: the effective degrees of freedom for identifying the level effect of EP depth are approximately 14, which severely limits power at the agreement level and motivates the shift to the within-firm product-composition strategy described in Section 5.

2.2 Variation in EP Depth: The Two Indices

This paper uses two independent measures of EP depth, which enables cross-validation and robustness checks. The indices differ in their coding methodology and scope but both aim to capture the substantive environmental ambition of China's agreements:

Dimension	World Bank DTA Index	TREND Dataset
Source / Reference	World Bank Deep Trade Agreements database (Hofmann-Osnago-Ruta)	Morin, Dür & Lechner (2018), Global Environmental Politics
Coverage	52 types of provisions; 'Environmental Laws' area used	286 types of EPs coded across 630+ PTAs worldwide
Index construction	Count of provisions in the Environmental Laws area; maximum across agreements for each destination-year	Count of environmental provisions; dedicated EP module
Granularity	Sub-indices: GreenLiberalization, EnforcementDSM, StandardsNonRegression, RegulatorySpaceExceptions, Assistance, Hardness_Share	Sub-indices: GreenMarketAccess, EnforcementDSM, RegulatorySpace, ClimateEnergy, BiodivForestsFisheries, Soft, Hard, Hardness_Share
Variation level	Destination × year (d,t)	Destination × year (d,t)
Advantage	Harmonised with broader DTA depth measures; enables TotalDepth control	More fine-grained EP taxonomy; independent coding — cross-validation

Both indices are pre-merged into the final analysis dataset at the destination-year level and available as columns in the FST file. The correlation between WB_EP_Depth and TREND_EP_Count is high but imperfect, which is why we estimate all main specifications with both indices and require qualitative agreement between the two for any finding to be considered robust.

2.3 The Identification Challenge Inherent in Level-Effect Designs

Both indices share a fundamental limitation for level-effect identification: EP depth switches from zero to a positive value exactly when the agreement enters into force, and remains positive thereafter. This means that in a standard difference-in-differences design, the EP depth variable is perfectly collinear with the agreement entry dummy. The coefficient on EP depth captures the joint effect of *everything* that changes when the agreement enters into force — not specifically the environmental provisions. With only ~14 agreement-level variation units, separating the EP effect from the general liberalization effect is infeasible at the level. This is the fundamental reason why the triple-difference on product composition is the paper's main identification strategy.

3. Data

3.1 Chinese Customs Transaction Data

The core dataset consists of Chinese export customs records for 2000–2015, obtained from the Chinese Customs General Administration and processed into a balanced panel of firm-product-destination-year observations. The raw data record every customs declaration, including exporter identity, HS6 product code, destination country, shipment value (USD), and quantity.

After cleaning, deduplication, and merging with the EP indices, the working dataset contains approximately **49.2 million observations**. It is stored as a compressed FST file (*final_dataset_pta_env_indices_compressed.fst*, approximately 14 GB on disk) enabling fast column-selective loading via the R *fst* package. The full dataset is never loaded into memory in a single R session; analysis scripts load only the columns required by the formula being estimated.

3.2 Outcome Variables

Variable	Label in Dataset	Description
Log export value	In_export	Natural log of total export value in USD for a firm-product-destination-year cell. Primary outcome. Captures the intensive margin of trade.
Log export quantity	In_export_qua	Natural log of export quantity (physical units as declared in customs). Complements the value outcome: if both move together, the effect is on volume; if value moves more, there is a quality/price component.
Log unit value	In_export_value	Natural log of unit value (USD per declared unit). Proxy for product quality and pricing. Note: unit values in raw customs data are noisy — trimming at the 1st/99th percentile within HS2-year is planned (Fase R1).

3.3 Treatment Variables

Treatment varies at the destination (d) × year (t) level. Both indices are available in the dataset as pre-merged columns:

Variable	Source	Range	Description
WB_EP_Depth	World Bank DTA	0 – ~8	Depth of environmental provisions in China's PTA with destination d, counting provisions in the Environmental Laws area. Zero before the agreement enters into force.
TREND_EP_Count	TREND database	0 – ~20	Count of environmental provisions from the TREND dataset. Independent coding; used as cross-validation for WB_EP_Depth.

Variable	Source	Range	Description
WB_TotalDepth	World Bank DTA	0 – ~50	Total PTA depth (all provisions, not just environmental). Used as a control to separate EP effects from general depth effects. STILL TO BE CONSTRUCTED from WB DTA Excel files (Script 06).

3.4 Product-Level Environmental Classification

The analysis requires classifying HS6 products by their environmental intensity. Two classifications are used (one available, one to be built):

Classification	Variable	Source	Status	Description
Green goods	env_good	OECD/APEC list	Available in dataset	Binary indicator: 1 if the HS6 product is on the combined OECD/APEC list of environmental goods (~142 HS6 codes, approximately 2.8% of world exports). Covers pollution-abatement equipment, clean energy goods, water treatment, air management equipment.
Dirty goods	dirty_p	Shapiro (2021) / ISIC	Script written (05); pending execution	Binary indicator: 1 if the product's industry is in the top quartile of CO2 intensity (emissions per USD of output). Source: Shapiro (2021) replication data (CO2 per industry), concordance ISIC → HS6 via R concordance package. Robustness: Mani-Wheeler (1998) industrial classification.

Products not classified as green or dirty are treated as neutral. In the triple-difference design, the coefficients on EP_depth:green_p and EP_depth:dirty_p measure the differential export response of green and dirty products relative to neutral products within the same firm-destination-year cell.

3.5 Pre-Computed Fixed-Effect Group Identifiers

A distinctive feature of the dataset is that all fixed-effect group identifiers are pre-computed as integer IDs in the FST file, avoiding the need to recompute them at estimation time (which would be slow at 49M observations). The available FE IDs are:

FE ID	Levels	Description
fpd	firm × product × destination	Captures the time-invariant characteristics of the bilateral relationship at the product level: buyer-seller relationships, product-specific comparative advantage, distance and institutional costs, product-market fixed effects.
fpt	firm × product × time	Captures what a given firm does with a given product across all destinations in a given year: firm productivity shocks, product-level supply shocks, firm-product-specific technology changes.

FE ID	Levels	Description
fdt	firm × destination × time	KEY for the triple-diff design. Captures everything that happens to a given firm in a given destination in a given year: PTA entry, tariff cuts, demand shocks, exchange rates. Absorbs the entire treatment event.
pt	product × time	Global demand shocks for a product: solar panel boom, commodity price cycles, global value chain changes.
pd	product × destination	Time-invariant bilateral comparative advantage at the product level.
dt	destination × time	Destination-year effects: importer-side macro shocks, income growth, PTA entry (for all firms).
pdt	product × destination × time	Most granular interaction: product-market-year shocks. Used in some robustness specs.
fp2dt	firm × HS2 × destination × time	Firm-chapter-destination-year: intermediate between fdt and fpt×dt.

3.6 Control Variables

Variable	Description	Status / Note
tariffs (ln_duty)	$\ln(1 + \text{MFN duty rate})$ on HS6 product p in destination d in year t.	AVAILABLE but MISCODED: this is the Most-Favoured Nation tariff, not the bilateral preferential rate applied to Chinese goods under the PTA. For PTA partner countries, the MFN rate does not decline after agreement entry. Replacing with preferential tariff from WITS TRAINS is Fase R2 priority.
tariffs_pref	$\ln(1 + \text{applied preferential tariff rate})$ — bilateral China-specific AHS.	TO BE BUILT via Script 04 (WITS TRAINS API). Will replace ln_duty in the main triple-diff specification.
ln_hhi_baci	Log of the Herfindahl-Hirschman Index for market concentration in product p × destination d.	Available. Controls for market structure at the product-destination level.
AD_pdt	Anti-dumping measure indicator: 1 if an anti-dumping measure is in force against China for product p in destination d in year t.	Available. Important confounder: anti-dumping actions may coincide with PTA signing and may be more frequent for specific product categories.

3.7 Data Still To Be Assembled

Data Input	Script	Source / Method	Priority
Preferential tariffs (tariffs_pref_pdt)	04_wits_pref_tariffs.R	WITS TRAINS database via SDMX API. Reporter = PTA partner countries, partner = CHN (156), tariff type = AHS (preferential applied). Syntax verified: rest/data/DF_WITS_Tariff_TRAINS/A.{rep}.156.{hs6}.AHS.ALL. Caches results by file to avoid redundant API calls. Validation: tariff should decline after agreement entry for PTA partners.	CRITICAL — without this, the tariff control is mis-specified
Dirty goods (dirty_p)	05_dirty_goods.R	CO2 intensity per ISIC industry from Shapiro (2021) replication package (Harvard Dataverse). Concordance ISIC → HS6 via R concordance package. Top quartile of CO2 intensity = dirty. Robustness: Mani-Wheeler (1998) classification of pollution-intensive industries.	CRITICAL — required for triple-diff
TotalDepth (non-environmental)	06_total_depth.R	WB DTA Excel files already in repository (DTA 1.0 and 2.0). Count all provisions excluding the Environmental Laws area. Merge at destination-year level same as EP indices.	HIGH — required to separate EP from general PTA depth

3.8 Dataset Scope and Limitations

- The dataset covers **export records only**. Import records and processing-trade regime flags are not available in the processed file, limiting GVC-type analyses such as imported input quality or backward GVC participation.
- The **ownership type** (state-owned vs. private vs. foreign-invested enterprise) is not available in the processed dataset. It may be recoverable from the raw Stata file (*final_dataset_pta.dta*); this would enable an important heterogeneity analysis (SOEs may respond differently to treaty commitments).
- **Hong Kong and Macao** are in the dataset but will be excluded from the main sample. Both are entrepôt hubs — Chinese exports recorded as 'to HK' are often re-exports to third countries. The CEPA agreements with HK and Macao are also structurally different from China's other PTAs (no standard environmental chapter). They will be included as a robustness check.
- The **HS concordance** across the three revision years within the 2000–2015 panel (HS 2002 → 2007 → 2012) is unverified. If codes are not concorded to a single vintage, the fpd and fpt fixed effects are artificially broken at revision boundaries. Script 02 will quantify this risk.

4. Diagnostic Findings from the Original Analysis

The project began with a standard gravity-style analysis — four fixed-effect structures estimated with OLS-HDFE, plus PPML and CEM robustness checks. A systematic audit of all 64 result tables (OLS + PPML × WB + TREND × 4–5 FE structures × full + CEM sample) revealed several important findings, summarized here. These diagnostics motivated the redesign described in Section 5.

4.1 The Level Effect: Monotone Attenuation Across Fixed-Effect Structures

The most important diagnostic finding is that the coefficient on EP depth (the 'level effect' — beta on WB_EP_Depth or TREND_EP_Count in the baseline specification) attenuates monotonically as fixed-effect saturation increases:

FE Structure	EP Depth Coef.	Apparent Significance	Clustering	Interpretation
fpd + year (least saturated)	Positive, ~0.03–0.05	2–3 stars on WB; 1–2 on TREND	~pdt (~2.9M 'clusters')	ARTEFACT. Clustering at pdt treats each firm-product-destination as an independent cluster — massively understates SEs. Treatment varies at (d,t) level, so correct clustering unit is country_code (~179 clusters, ~25 treated).
fpt + pd	Positive, ~0.01–0.02	Reduced significance (1 star or less)	~dt	Adding more FEs absorbs part of the selection. Still positive but smaller. dt clustering is appropriate here.
fpt + fpd (preferred main spec)	Near zero (~0.002)	Not significant	~country_code	The fpd FE absorbs the time-invariant level of the bilateral relationship, including most of the selection. Coefficient near zero with correct clustering.
fpd + pt	Near zero (~0.001)	Not significant	~country_code	Alternative absorbing product-time shocks. Same conclusion: level effect is a precise null with correctly-saturated FEs.

This monotone pattern is the **signature of positive selection**: destinations that sign deeper PTAs with China tend to be richer, faster-growing markets with more sophisticated demand. Firms that export to these markets would have increased their exports regardless of the EP content. The fpd fixed effect absorbs this time-invariant selection, removing the spurious correlation. The stars in the least-saturated specification were entirely driven by clustering at the wrong level — this is a clean example of the Bertrand-Duflo-Mullainathan (2004) under-clustering problem.

This diagnostic finding is itself a publishable contribution: it provides the first firm-level evidence that the aggregate EP effects documented by Brandi et al. (2020) are likely driven by selection rather than causal identification. The 'ladder table' (OLS_Ladder_FE.tex) documenting this attenuation will

be a key exhibit in the paper.

4.2 The Interaction Effect: A Credible Signal

One coefficient proved more stable across specifications: the interaction of EP depth with the green good indicator ($WB_EP_Depth \times env_good$). Across several specifications, this was:

Specification	EP $\times$ env_good Coefficient	Standard Error	Notes
fpt + fpd, WB, value, no controls	+0.0009	~0.0004	Survives move to country_code clustering. Positive and stat. significant.
fpt + fpd, WB, quantity, no controls	+0.0010	~0.0004	Similar magnitude. Effect on volume, not just price.
fpt + fpd, TREND, value, no controls	+0.0011	~0.0005	TREND index confirms the direction. Slightly larger.
fpt + fpd, WB, value, with controls	+0.0007	~0.0005	Attenuates slightly with controls but remains positive. Borderline.

This positive interaction — deeper EP agreements are associated with differentially higher green exports — is the signal that the triple-difference design is intended to test rigorously. The magnitude is economically modest but the sign consistency across WB and TREND indices and across value/quantity outcomes is encouraging. The key question is whether this survives the full triple-diff specification with *dirty_p* as the counterfactual product group.

4.3 Other Diagnostic Issues Identified

- **MFN tariff control is mis-specified.** The duty variable contains MFN tariff rates, not bilateral preferential rates. For PTA partner countries, the MFN rate shows no systematic decline after agreement entry (confirmed by time-series plots). This means the tariff coefficient — while the most stable in the original tables — is measuring the effect of the *external* tariff structure on export composition, not the PTA tariff preference. This is an important mis-specification for the mechanism: part of what we attribute to EP depth may be the correlation between EP provisions and deep tariff preferences.
- **PPML internally inconsistent.** PPML estimates show sign reversals across outcomes (value positive, quantity sometimes negative) and across FE structures. The fpt-only PPML specification is a clear outlier with implausibly large coefficients — likely due to imperfect collinearity between the FE structure and the treatment variable when no bilateral relationship FE is included. PPML on unit values makes no conceptual sense (unit value is not a count/flow) and will be dropped. The two PPML robustness strategies to keep are described in Section 7.3.
- **CEM matching is cosmetically weak.** 25 treated countries, 3 matching covariates (log GDP per capita, log GDP, year of agreement entry). L1 imbalance improves modestly from 0.788 to 0.652, but the standardized mean difference on log GDP per capita *worsens* after matching. With country-level treatment and 25 treated units, kernel matching on 3 covariates is cosmetic. The robust alternative is described in Section 7.4.

- **Hong Kong and Macao contamination.** CEPA with HK and Macao is included in the treated group in the original analysis. Given HK's role as an entrepôt re-export hub and the structurally different nature of CEPA, these two destinations should be excluded from the main sample.
- **Anti-dumping as confounder.** The AD_pdt variable is available but used inconsistently across the original specifications. Anti-dumping measures imposed on Chinese goods in PTA partner countries may correlate with both agreement signing and product composition — they are a natural confounder to include in the triple-diff.

5. Redesigned Empirical Strategy

5.1 The Core Identification Problem and Its Solution

The fundamental identification challenge is that EP depth — both WB_EP_Depth and TREND_EP_Count — varies only at the destination $\times$ year level. It switches from zero to a positive value exactly when the agreement enters into force and remains positive thereafter. This means that EP depth is perfectly collinear with the *event* of PTA entry: any coefficient on EP depth alone estimates the joint effect of having a deep PTA (tariff cuts, investment provisions, trade facilitation, EP clauses) rather than the specific effect of environmental provisions.

With only ~14 effective agreements (the effective variation units), disentangling the EP contribution from the general liberalization effect is infeasible at the level-effect stage. Any serious referee at JIE or JEEM would reject the level effect as identification for EP-specific causal effects.

The solution is to abandon level-effect identification and instead exploit the **differential effect across products of different environmental intensity**. The hypothesis is that EP depth *specifically* reshapes export composition — green goods should increase relative to dirty and neutral goods — because environmental provisions create incentives for clean production and market access for environmental goods that do not apply uniformly across the product range. A general deep PTA should not differentially favor green over dirty products (unless it specifically includes green market access provisions — which is exactly what we are testing). The TotalDepth control separates these two channels.

5.2 The Triple-Difference Specification

The main estimating equation is:

$$\ln_export(f, p, d, t) = \beta_{EP} \cdot EP(d, t) \cdot green_p + \beta_{EP} \cdot EP(d, t) \cdot dirty_p + \beta_{TD} \cdot TotalD(d, t) \cdot green_p + \beta_{TD} \cdot TotalD(d, t) \cdot dirty_p + \gamma \cdot tariffs_pref(p, d, t) + \delta \cdot AD(p, d, t) + \alpha(f, p, d) + \alpha(f, d, t) + \alpha(p, t) + \epsilon(f, p, d, t)$$

Estimated in R using fixest:

```
feols(ln_export ~ EP_depth:green_p + EP_depth:dirty_p + TotalDepth:green_p +
      TotalDepth:dirty_p + tariffs_pref + AD_pdt | fpd + fdt + pt, cluster =
      ~country_code, data = dt)
```

The key coefficients of interest are β_{EP} (EP $\times$ green) and β_{EP} (EP $\times$ dirty). The hypothesis consistent with EPs being effective is $\beta_{EP} > 0$ and $\beta_{EP} < 0$. β_{TD} and β_{TD} are controls for the general depth channel; if $\beta_{TD} \approx \beta_{TD}$ and $\beta_{TD} \approx \beta_{TD}$, it is the general depth rather than the EP content that drives the composition effect. If $\beta_{EP} \neq \beta_{EP}$ (specifically $\beta_{EP} > \beta_{EP}$), this is evidence that it is specifically the environmental provisions that matter.

5.3 Fixed Effects: What Each Absorbs

Fixed Effect	Notation	Levels of Variation Absorbed	Why It Matters
Firm × Product × Destination (fpd)	$\alpha(f,p,d)$	Time-invariant characteristics of the firm-product-destination triple: buyer-seller relationships, product-specific comparative advantage, iceberg transport costs, product-market entry costs.	Removes the level of the bilateral relationship, which is the primary channel of selection (richer markets get more exports and deeper PTAs simultaneously).
Firm × Destination × Year (fdt)	$\alpha(f,d,t)$	EVERYTHING that varies for firm f in destination d in year t : PTA entry into force, all tariff cuts, all investment provisions, bilateral exchange rate, destination income growth, the EP depth level itself.	The KEY identifying FE. With fdt in the model, the only residual variation is across products within the same firm-market-year cell. The EP level effect is absorbed. β_{green} and β_{dirty} are identified only from product heterogeneity.
Product × Year (pt)	$\alpha(p,t)$	Global demand shocks for product p in year t : the solar panel boom, global commodity supercycle, GVC restructuring shocks that affect all exporters of product p .	Ensures that green products are not trending globally for reasons unrelated to Chinese PTAs.

5.4 Identification

With the fdt fixed effect, the coefficients β_{green} and β_{dirty} are identified from variation *within* a firm-destination-year cell, *across* products of different environmental intensity. Formally, we are comparing:

- Do firms export *relatively more green* goods (and *relatively less dirty* goods) to destinations with deeper EP agreements, compared to what those same firms export to the same destinations in years before the agreement entered into force (or compared to their exports of neutral products)?

This is the Rajan-Zingales (1998) difference-in-differences across industries applied to environmental provisions. The identifying assumption is that, in the absence of EP provisions, the ratio of green to neutral exports would not systematically increase more in EP-deep destinations than in other destinations — conditional on all the variation absorbed by the three FEs.

This assumption is more credible than the level-effect parallel trends assumption because: (a) the fdt FE absorbs any differential trend in the level of bilateral exports; (b) the product-specific composition is determined by supply-side factors (technology, input costs) and demand-side factors (importer preferences) that do not obviously differentially affect green vs. dirty products for EP-deep destinations specifically.

5.5 Pre-Trend Analysis and Event Study

To validate the identification assumption, we estimate an event study around the entry into force of each agreement, looking for differential pre-trends in the green vs. neutral product ratio:


```
feols(ln_export ~ i(rel_time, green_p, ref = -1) + i(rel_time, dirty_p, ref = -1)
| fpd + fdt + pt, cluster = ~country_code)
```

where `rel_time` is years relative to agreement entry into force, ranging from -5 to +5 (saturated dummies). We expect: (a) pre-treatment coefficients on `i(rel_time, green_p)` to be near zero and not systematically increasing (parallel trends in composition); (b) post-treatment coefficients to diverge positively for green and negatively for dirty products if EPs are effective.

Since agreements entered into force at different times across the sample period (staggered design), we use `sunab()` from the `fixest` package to implement the Callaway-Sant'Anna (2021) / Sun-Abraham (2021) approach, which avoids the negative-weighting problem identified by Goodman-Bacon (2021) for heterogeneous-timing TWFE estimators.

5.6 Inference Strategy: Three Layers

With only ~14 effective treatment clusters (one per agreement), standard asymptotic cluster-robust SEs may be unreliable. We report three layers of inference for the main specification:

Layer	Method	Implementation in R	Rationale
1 — Baseline	Cluster-robust SEs at <code>country_code</code> level	<code>cluster = ~country_code</code> in <code>feols()</code> ; 179 clusters, ~25 treated	Asymptotically valid; reported as primary. Accounts for within-destination serial correlation. With 179 clusters total and ~25 treated, asymptotic theory is borderline reliable.
2 — Bootstrap	Wild Cluster Bootstrap ($B = 9,999$)	<code>fwildclusterboot::boottest()</code> on the fitted <code>feols</code> object	More reliable than asymptotic clustering with few treated clusters (Cameron, Gelbach & Miller 2008). Imposes the null on the treatment variable; p-values from the bootstrap distribution.
3 — Permutation	Permutation inference: randomize EP depth over ~14 agreements	1,000 random reassignments of EP depth vector across ~14 agreements; PTA timing fixed; compute t-stat each draw; exact p-value from distribution	Exact p-value for the null that EP content is random, holding fixed the timing and existence of agreements. Tests specifically whether environmental content — not the agreement itself — drives the composition effect.

We require robustness to all three inference methods for a result to be considered credible. In the paper, all three p-values will be reported in parentheses below the coefficients in the main table.

6. Related Literature

6.1 Environmental Provisions in Trade Agreements

The literature on EPs in trade agreements has developed rapidly, largely enabled by the TREND database (Morin et al. 2018), which codes 286 types of EPs across 630+ PTAs. Key studies and their relevance:

Paper	Journal / Outlet	Main Finding	Relevance to This Paper
Brandi, Schwab, Berger & Morin (2020)	World Development	EPs in PTAs reduce the dirty export share and increase the green export share for developing countries. Country-pair-year gravity panel, 1984–2016. Controls: GDP, distance, colonial ties, RTA depth.	Most direct predecessor. Our paper extends to firm level, focuses on China, and corrects the identification by absorbing the PTA entry through fdt FEs. The level-null against Brandi is itself a contribution.
Abman, Lundberg & Ruta (2024)	JEEA (22:6)	RTAs with forest/biodiversity provisions fully offset the deforestation caused by trade liberalization (satellite data). Effect is heterogeneous: only provisions with DSM (dispute settlement mechanisms) are effective.	Closest top-journal precedent for EP causal identification. Key insight: enforcement matters (hard vs. soft provisions). We incorporate the hard/soft distinction in our sub-index heterogeneity analysis.
Morin, Dür & Lechner (2018)	Global Environmental Politics	Introduces the TREND dataset; documents the rapid proliferation of EPs across PTAs and the heterogeneity in their content. Shows that EP density increases with overall PTA ambition.	Data source for the TREND_EP_Count index. The correlation between EP density and overall depth motivates the TotalDepth control (Fase R2).
Neri-Laine, Orefice & Ruta (2023)	CESifo WP	Deep PTA provisions boost large-firm exports but hurt small firms' market entry. Firm-level EDD data. PPML with saturated FEs.	Design template for our firm-size heterogeneity analysis (Fase R4). Also illustrates that aggregating across firm sizes masks heterogeneity.

6.2 Trade, Pollution, and the Pollution Haven Hypothesis

Paper	Journal	Main Finding	Relevance
Cherniwchan (2017)	J. Intl. Econ.	NAFTA tariff cuts caused within-plant reductions in PM10 and SO2 from US manufacturing. ~2/3 of the total US manufacturing emission decline 1994–98 is attributable to NAFTA. Demonstrates the technique effect at the micro level.	Template for micro-level trade-environment evidence using differential product/industry exposure to the same agreement. Justifies the triple-difference design: the same logic — differential response across dirty vs. clean industries — applied to EP provisions.
Shapiro (2021)	QJE	Trade policy is systematically biased: tariffs are lower on dirty (upstream) industries globally, amounting to an implicit CO2 subsidy of \$550–800bn/year. Driver: upstream location of dirty industries + lobbying.	Source for the dirty_p measure (CO2 intensity per ISIC industry). Warning: preferential tariff cuts may be deeper for dirty goods — must be controlled in the triple-diff to avoid spurious β estimates.
Copeland, Shapiro & Taylor (2022)	Handbook of Intl. Econ.	Comprehensive survey: theoretical framework decomposing trade effects into scale ($\uparrow$ pollution via output), composition (shift toward cleaner industries), and technique effects (cleaner production at given output). Policy can affect all three margins.	Theoretical framework for the paper. EP mechanisms: green market access affects composition margin; standards affect technique margin; both affect scale via trade volume.
Dechezleprêtre & Sato (2017)	REEP	Review: pollution haven effects exist but are modest; firm-level data reduce omitted variable bias substantially relative to industry-level studies.	Motivates firm-level analysis. Provides context for expected effect sizes.
Mani & Wheeler (1998)	J. Env. Dev.	Identifies pollution-intensive industries for the 'dirty goods' classification: iron/steel, non-ferrous metals, industrial chemicals, pulp/paper, non-metallic minerals.	Robustness classification for dirty_p alongside Shapiro (2021).

6.3 Identification and Econometric Methodology

Paper	Journal	Main Finding	Relevance
Santos Silva & Tenreyro (2006)	Rev. Econ. Stat.	Log-linear gravity models are inconsistent under heteroskedasticity. PPML is consistent and has better small-sample properties.	Justifies PPML robustness checks. Firm-level PPML on positive flows corrects heteroskedasticity; aggregate PPML handles zeros.

Paper	Journal	Main Finding	Relevance
Goodman-Bacon (2021)	J. Econometrics	With staggered treatment timing, TWFE estimates are variance-weighted averages of 2×2 DiDs, some with negative weights (using later-treated units as controls for earlier-treated units).	Motivates the sunab() approach in the event study. With 14 agreements at different entry times, standard TWFE event studies are unreliable.
Callaway & Sant'Anna (2021)	J. Econometrics	Proposes group-time average treatment effects that avoid the negative-weighting problem. Requires 'not-yet-treated' or 'never-treated' control groups.	Implemented in Fase R5 as a robustness check on the binary treatment version. Requires splitting by agreement (group = entry year).
de Chaisemartin & D'Haultfoeuille (2020)	American Econ. Rev.	Identifies conditions under which TWFE estimates can be negative even when all unit-level treatment effects are positive. Proposes heterogeneity-robust DiD.	Same motivation as Goodman-Bacon. dCDH estimator as additional robustness check.
Cameron, Gelbach & Miller (2008)	Rev. Econ. Stat.	Wild cluster bootstrap provides more reliable inference than asymptotic cluster-robust SEs when the number of clusters is small.	Justifies B=9,999 bootstrap. With ~14 treatment clusters, WCB is the primary inference method alongside permutation.
Bertrand, Duflo & Mullainathan (2004)	QJE	DiD SEs must account for within-group serial correlation; ignoring this massively understates SEs and overstates significance (spurious 60–70% rejection rate at 5% nominal level).	Motivates correct clustering at country_code. The spurious stars in the fpd+year spec (clustered at pdt) are precisely this problem.
Larch, Shikher & Yotov (2025)	Review of Intl. Econ.	15 recommendations for applied gravity: PPML with zeros; complete bilateral pairs; time-varying FEs; within-country references; directional trade costs.	Informs aggregate PPML specification: rectangular grid with zeros, FE pd + pt + dt, as recommended by Larch et al.

7. Current Status of the Analysis

The project is currently in **Fase R0**: completing the diagnostic inference exercise (re-estimation with correct clustering + wild bootstrap + ladder table) before moving to the main triple-difference analysis. The following tables document the completion status of all tasks across the roadmap phases.

7.1 Fase R0 — Computational Progress: 01_inference_fix.R

Script 01 re-estimates OLS models across four FE structures with uniform country_code clustering, wild bootstrap (B=9,999), and the ladder table. Each model on the fpt+fpd and fpd+pt structures takes approximately 50–60 minutes at 10 threads (the regression involves ~49M observations with dual high-dimensional fixed effects). Status at time of writing:

FE Structure	WB — No Int	WB — Interactions	TREND — No Int	TREND — Int
fpd + year	6/6 DONE	6/6 DONE	6/6 DONE	6/6 DONE
fpt + pd	6/6 DONE	6/6 DONE	6/6 DONE	6/6 DONE
fpt + fpd	6/6 DONE	6/6 DONE	6/6 DONE	RUNNING
fpd + pt	PENDING	PENDING	PENDING	PENDING

Note: 'No Int' = specification without interaction terms (level effect only). 'Interactions' = specification with EP_depth × env_good and EP_depth × TotalDepth interaction terms. Each cell = 6 models (3 outcomes × {baseline, controls}). Total: 48 models across all FE structures. Currently ~36/48 complete. Remaining estimated computation time: 1–2 days.

After all 48 base models complete: wild cluster bootstrap (B=9,999) will run on the fpt+fpd specification (the 'preferred' structure) for the key interaction coefficients. Then the ladder table (OLS_Ladder_FE.tex) will be generated programmatically from the stored RDS files.

7.2 Full Roadmap: All Phases

Phase	Status	Description	Key Deliverable
R0 — Close Diagnostic	IN PROGRESS	Complete 01_inference_fix.R: remaining OLS models (fpd+pt section, ~24 models), wild cluster bootstrap (B=9,999 on fpt+fpd), ladder table. One R process at a time to avoid the OpenMP allocator crash.	bootstrap_summary.csv confirming null on level; OLS_Ladder_FE.tex with 4 rows (monotone attenuation).

Phase	Status	Description	Key Deliverable
R1 — Data Hygiene	PENDING	Run 02_data_hygiene_audit.R. Five checks: (1) HS concordance across 2002/2007/2012 revisions — CRITICAL, could invalidate existing results; (2) treatment table: 14 agreements, switch dates, EP depth values; (3) HK+Macao weight in the treated sample; (4) UV outlier trimming 1%/99% within HS2-year; (5) firm consistency check around 2004 (WTO liberalisation of trading rights).	Diagnostic report in New/Output/Diagnostics/. Decision documented: concordance needed yes/no.
R2 — New Data	PENDING (scripts written)	Run 04 (WITS tariffs), 05 (dirty goods), 06 (TotalDepth). Scripts written and verified; need to execute. 04 can run in parallel to R1 (only requires internet access, no FST). 05 and 06 are fast (small datasets).	tariffs_pref_pdt.csv, dirty_goods_hs6.csv, total_depth_dt.csv — all merge-validated.
R3 — Main Triple-Diff	PENDING	Run 07_triple_diff.R. Three sections: (A) main triple-diff estimates — 3 outcomes × {WB, TREND} with all three inference methods (cluster + WCB + permutation); (B) event study with sunab() for staggered design; (C) 1,000-draw permutation test. Also: PPML firm-level on positive flows (same triple-diff spec with feipois) + PPML aggregate pdxt with zeros.	Table_TripleDiff_Main.tex; EventStudy plot; permutation_pvalues.csv.
R4 — Margins and Mechanisms	PENDING	Extensive margin: n firms / n products green per dxt cell. Within-firm reallocation: green share of multi-product firms' export basket toward d (identified by fdt FE — the within-firm variation is key). EP sub-index heterogeneity: GreenMarketAccess, EnforcementDSM, Hard/Soft. Firm size heterogeneity (terziles of total export value).	Extensive margin table; within-firm green share table; sub-index table.
R5 — Robustness	PENDING	Leave-one-out per agreement (14 specifications). Exclude ASEAN bloc. Include/exclude HK+Macao. Not-yet-treated only as control group. Callaway-Sant'Anna on binary treatment version. Synthetic DiD on destination-level green export share. UV trimmed vs. raw. WB vs. TREND in all specs.	Appendix tables; discussion of sensitivity.
R6 — Framing and Writing	PENDING	Framing decision after R3 results are available. Descriptive section: position China's EP agreements in the global TREND distribution (shallow vs. deep; timeline; EU/US comparison). Reduce from 48+ tables to 6–8 main tables + structured appendix. Target journal selection.	Paper draft; target journal decision.

7.3 Scripts in the New/ Directory

Script	Phase	Status	Description
01_inference_fix.R	R0	RUNNING	Four FE structures × {WB, TREND} × {No-Int, Int} × {baseline, controls}, all clustered at country_code. Uses callr::r() to serialize each section as a subprocess (avoids OpenMP/FST allocator conflict on Windows). Wild cluster bootstrap section. Ladder table generation.

Script	Phase	Status	Description
02_data_hygiene_audit.R	R1	WRITTEN — ready	HS concordance audit (count dying/new codes at each revision boundary), treatment table (14 agreements × dates/depth), HK+Macao export weight, UV trimming flags, firm consistency check around 2004.
04_wits_pref_tariffs.R	R2	WRITTEN — ready	Preferential tariff download from WITS TRAINS via SDMX API. Loops over PTA partner countries, downloads AHS rates for China (partner 156), HS6 level, 2000–2015. Syntax verified against WITS SDMX endpoint. File-based cache to avoid redundant API calls.
05_dirty_goods.R	R2	WRITTEN — ready	Dirty goods classification. Loads Shapiro (2021) CO2 intensity data (ISIC industry level). Concordance ISIC→HS6 via R concordance package. Top quartile = dirty. Robustness: Mani-Wheeler (1998) industry list. Output: dirty_goods_hs6.csv with dirty_p indicator.
06_total_depth.R	R2	WRITTEN — ready	TotalDepth (all PTA provisions excluding environmental). Reads WB DTA Excel files from repository. Constructs total provision count and non-environmental count. Merges at destination-year level. Validation: correlation with existing WB_EP_Depth.
07_triple_diff.R	R3	WRITTEN — ready	Full triple-difference: Section A (main OLS estimates, 3 outcomes × {WB,TREND}, all three inference methods), Section B (event study with sunab()), Section C (1,000-draw permutation inference). Requires: dirty_p, TotalDepth, tariffs_pref from Fase R2.

8. Key Pending Decisions and Open Questions

8.1 Critical Decision: HS Concordance

This is the most urgent unresolved issue. The panel covers 2000–2015, crossing three HS revision years: 2002 (HS1996 → HS2002), 2007 (HS2002 → HS2007), and 2012 (HS2007 → HS2012). At each revision, a subset of HS6 codes are split, merged, or renumbered. If the customs data have not been concorded to a single HS vintage, two problems arise:

- The firm-product-destination fixed effects (fpd) are *artificially broken* at revision boundaries: a product that existed continuously under one code is treated as two different products before and after reclassification. This introduces spurious entry and exit in the panel.
- The `env_good` indicator — which is mapped to a specific HS vintage — is *incorrectly assigned* for portions of the sample: a green product may be classified as neutral before a revision, or vice versa.

The audit script (02_data_hygiene_audit.R) will quantify the share of HS6 codes that change across each revision boundary and the fraction of total export value they represent. Decision rule:

Scenario	Action Required
< 1% of export value affected by revision discontinuities	No action required. Document and note as minor limitation.
1–3% of export value affected	Partial concordance: remap affected codes only. Re-verify <code>env_good</code> mapping. Note as limitation with sensitivity check excluding revision-boundary years.
> 3% of export value affected	Full concordance required: rebuild pipeline Steps 2–3 using UNSD concordance tables (HS1996–HS2002–HS2007–HS2012). This is several weeks of work but is essential for publishability. New FST file will be required.

8.2 Framing Decision: After Fase R3

The ultimate framing and target journal depend on the triple-difference results. Two scenarios and their implications:

Scenario	Result Pattern	Framing	Target Journal
A — Signal survives	$\beta_{\blacksquare}(\text{EP} \times \text{green}) > 0$ and $\beta_{\blacksquare}(\text{EP} \times \text{dirty}) < 0$, both significant under all three inference methods. Event study: no pre-trends, clear post-treatment divergence in green/dirty ratio. Permutation $p < 0.05$.	Title: 'Do Environmental Provisions Green Exports? Firm-Level Evidence from China'. Headline: triple-diff composition effect + within-firm green reallocation (Fase R4). Level null as methodological contribution against Brandi (2020). The within-firm reallocation result would be novel and compelling.	JIE or JEEM (full paper). Potentially REStud/AEJ:EP if within-firm result is strong.

Scenario	Result Pattern	Framing	Target Journal
B — Precise null	Neither β_{green} nor β_{dirty} survives the more conservative inference methods. Permutation $p > 0.10$. Event study shows no differential pre/post pattern.	Title: 'Are Environmental Provisions in Chinese PTAs Effective? A Micro-Level Null'. Contribution: first firm-level test; ladder documenting selection in aggregate studies; methodological critique of Brandi (2020) and ALR (2024); permutation test ruling out EP content as driver.	World Development or JEEM (short paper). Still publishable as a methodological contribution.
C — Mixed	β_{green} (green) survives but β_{dirty} (dirty) does not, or vice versa. Or: survives WCB but not permutation.	Framing depends on which sub-result is more robust. Focus on the green market access sub-index if GreenMarketAccess drives β_{green} . Discuss dirty goods identification uncertainty as a caveat.	JEEM or similar environmental economics journal.

8.3 PPML Strategy

Two PPML estimations are planned, serving different purposes:

PPML Type	Specification	Purpose	R Implementation
Firm-level PPML (positive flows only)	Same triple-diff as OLS: <code>fepois(ln_export_raw ~ EP:green + ... fpd + fdt + pt)</code> On strictly positive export flows only.	Corrects for heteroskedasticity in the intensive margin (Santos Silva & Tenreyro 2006). Directly comparable to OLS as robustness check. Firms $\times$ products $\times$ destinations with zero exports in a given year are excluded.	<code>fepois()</code> $\rightarrow$ <code>fepois()</code> drop-in. Note: <code>fepois</code> does not accept log outcome; use raw export value. R^2 is meaningless in PPML — will not be reported.
Aggregate PPML (with zeros)	Collapse to product $\times$ destination $\times$ year: <code>fepois(export_raw ~ EP:green + ... pd + pt + dt)</code> On rectangular grid including zero-trade cells.	Captures the extensive margin: whether EP depth induces new product-market pairs to start trading. Comparable to Brandi (2020) and Larch et al. (2025). The 'true' PPML as recommended in the gravity literature.	Aggregate to <code>pdyt</code> grid (~14M cells). <code>fepois</code> with FE <code>pd + pt + dt</code> . Large but feasible computation.

Note: PPML on unit values (`ln_export_value`) makes no conceptual sense — unit value is not a count/flow variable and PPML does not apply. All original PPML specifications on unit values will be dropped.

8.4 CEM: Appendix or Removal?

The original CEM matching is methodologically weak:

- 25 treated countries, 3 matching covariates (log GDP, log GDP per capita, year of entry). At the country level with limited degrees of freedom, kernel matching on 3 covariates is cosmetic.
- L1 imbalance improves from 0.788 to 0.652 — modest. More importantly, the standardized mean difference on log GDP per capita *worsens* after matching, indicating that CEM creates imbalance on some important dimensions while improving others.
- The matched control sample retains only a subset of the original control observations, reducing efficiency without credible identification gains.

The plan is to **demote CEM to an appendix** (or remove entirely) and replace the robustness strategy with: (a) not-yet-treated destinations as the control group; (b) synthetic DiD at the destination level on the green export share; (c) Callaway-Sant'Anna on the binary treatment version of EP depth (above/below median EP depth conditional on having a PTA).

8.5 Zotero Collection: Pending Items

Four key papers should be added to the Zotero Paper_PTA collection but are currently missing (Zotero was in local-only mode when last checked):

Paper	DOI	Priority	Why Missing
Abman, Lundberg & Ruta (2024) JEEA	10.1093/jeea/jvae023	CRITICAL — closest competitor	Zotero add-by-DOI failed in previous session (local-only mode).
Shapiro (2021) QJE	10.1093/qje/qjaa042	CRITICAL — dirty_p source	Same issue.
Cherniwchan (2017) JIE	10.1016/j.jinteco.2017.01.005	High	Same issue.
Copeland, Shapiro & Taylor (2022) Handbook	10.1016/bs.hesint.2022.02.002	High	Same issue.

9. Planned Heterogeneity Analyses

Once the main triple-difference results are established (Fase R3), the following heterogeneity analyses will be conducted. These are organized by what type of additional variation they exploit.

9.1 Extensive Margin of Green Exports

The main triple-difference operates on the intensive margin: export value/quantity conditional on positive trade. The extensive margin asks: does EP depth induce firms to start exporting green goods to a destination, or induce new products to enter new markets? We will estimate:

- **Firm-level extensive margin:** number of distinct firms that export at least one green product to destination d in year t . Regression of $\log(n_firms_green_dt)$ on EP_depth_dt with FE dt (or destination + year + their interaction). Compare growth in green exporters to growth in dirty/neutral exporters in treated vs. control destinations.
- **Product-level extensive margin:** number of distinct HS6 green products exported to d in year t . Same design.
- **Product entry:** for green products p not previously exported to d by firm f , does the year of first export shift toward the agreement entry year?

9.2 Within-Firm Green Share Reallocation

This is potentially the strongest result in the paper, and novel relative to all prior literature. For multi-product firms (which account for most of Chinese exports by value), does the *share of green products in the firm's export basket toward destination d* increase after the agreement enters into force?

Specification (Rajan-Zingales applied within multi-product firms):

$$green_share(f,d,t) = \beta * EP_depth(d,t) + FE(f,d) + FE(f,t) + \epsilon$$

where $green_share(f,d,t) = [\text{value of green exports from } f \text{ to } d \text{ in } t] / [\text{total exports from } f \text{ to } d \text{ in } t]$. The FE $f \times d$ absorbs the firm-destination-specific level; the FE $f \times t$ absorbs the firm-year trend in green specialization. The coefficient β identifies whether a firm shifts its product mix toward green goods specifically in destinations where it faces deeper EP agreements, controlling for any general trend in the firm's green intensity.

This is identified from firms that export to *both* EP-deep and EP-shallow destinations simultaneously, allowing a within-firm cross-destination comparison. No such test has been done in the literature.

9.3 EP Sub-Index Heterogeneity

Not all types of environmental provisions are equally likely to affect trade composition. Theory suggests a hierarchy:

Sub-Index	Mechanism	Expected Relevance	Available in Dataset?
GreenMarketAccess (TREND) / GreenLiberalization (WB)	Provisions specifically reducing tariffs and non-tariff barriers for environmental goods. Directly incentivizes green exports.	HIGHEST — this is the most direct trade-composition mechanism. If this sub-index drives $\beta_{\text{■}}$, the interpretation is clean market access.	Yes (both indices)
EnforcementDSM (both indices)	Provisions establishing dispute settlement mechanisms for environmental obligations. Abman et al. (2024) find this is the effective provision type.	HIGH — enforceable provisions with DSM are more binding. Hard provisions without DSM may be cheaper to circumvent.	Yes (both indices)
Hard vs. Soft obligations (Hardness_Share)	Share of provisions that are binding ('shall') vs. aspirational ('should', 'may'). Binding provisions create legal obligations.	MODERATE — consistent with ALR (2024) finding that hard provisions are what matters for deforestation.	Yes (as share; requires construction of hard-only sub-index)
StandardsNonRegression (WB) / RegulatorySpace (TREND)	Provisions preventing countries from weakening environmental standards to attract trade. Pollution haven suppression mechanism.	MODERATE — directly relevant to dirty goods: if firms cannot exploit lax environmental standards, dirty exports may be constrained.	Yes (WB has this; TREND equivalent available)

9.4 Firm-Size Heterogeneity

Following Neri-Laine, Orefice & Ruta (2023), we expect heterogeneous effects by firm size: large, established exporters may more easily adapt to EP-driven demand for green goods (they already have compliance infrastructure), while small firms may face higher fixed costs of product adaptation. We split the sample into terciles by total export value (or employment, if available) and estimate the triple-diff separately:

- Large firms: above 66th percentile of total annual exports.
- Medium firms: 33rd–66th percentile.
- Small firms: below 33rd percentile.

We expect $\beta_{\text{■}}$ to be larger in magnitude for large firms, consistent with the Neri-Laine et al. finding and with the general literature on firm-size and regulatory compliance costs.

10. Technical Infrastructure

10.1 Data Pipeline

The data pipeline runs in four sequential steps, combining R and Stata. The pipeline runs once to build the analysis dataset; all subsequent analysis runs only from the final FST file. Original datasets are read-only; all new work goes into the New/ directory.

Step	Language	Script	Input	Output
0	Stata	WB_Dataset_Conversion.do	WB DTA Excel files	WB_DTA.dta — World Bank DTA indices in Stata format
1	R	1_Build_Final_PTA_EP_Dataset.R	WB_DTA.dta + TREND raw data + country codes	Merged_TREND_WB_Indices_Only.dta — both indices at destination-year level
2	Stata	2_Build_Final_PTA_EP_Dataset.do	Chinese customs Stata file + Merged indices	final_dataset_pta_env_indices_compressed.dta (~14 GB)
3	R	3_Build_Final_PTA_EP_Dataset.R	The .dta from Step 2	final_dataset_pta_env_indices_compressed.fst — fast column-selective R format

Steps 0 and 2 are run interactively in Stata. Steps 1 and 3 are R scripts. The FST format (R package `fst`) enables sub-second access to any column subset of the 49M-row dataset without loading the full file into memory. All analysis scripts use `read_fst(path, columns = needed_cols)`.

10.2 The Shared Utility Library: `pta_functions.R`

All analysis scripts source `Code/Analysis/pta_functions.R`, which provides the core infrastructure for model estimation, caching, and table generation. Key functions:

Function	Purpose	How It Works
<code>run_block(block_name, formulas, ...)</code>	Runs a named block of fixest formulas with RDS caching.	For each formula in the block, checks if the corresponding RDS file already exists in <code>New/Output/OLS/</code> . If so, skips; if not, calls <code>estimate_model()</code> and saves the result. Enables incremental computation — a crashed job can restart from the last completed model.
<code>estimate_model(formula, fst_path, ...)</code>	Estimates a single model from the FST file, loading only required columns.	Calls <code>load_formula_data()</code> to parse the formula and identify needed columns. Loads only those columns from FST. Runs <code>feols()</code> or <code>fepois()</code> . Returns a stats object (not the full model, to save memory).

Function	Purpose	How It Works
<code>load_formula_data(formula_str, fst_path)</code>	Parses a fixest formula string and loads only necessary columns.	Regex-parses the formula to extract variable names. Calls <code>read_fst(fst_path, columns = needed_cols)</code> . Returns a <code>data.table</code> .
<code>make_table(rds_list, ...)</code>	Generates LaTeX regression tables from stored model stats objects.	Loads the list of RDS files, assembles coefficients + SEs into a matrix, and outputs a LaTeX table using the project's standard format.

10.3 Concurrency and the OpenMP / FST Crash

A critical technical constraint on Windows: running two or more R processes that *simultaneously* load the FST file causes an OpenMP allocator conflict that terminates both processes with the error `*** recursive gc invocation`. This occurs even at 4 threads per process, and is specific to the interaction between the FST package (which uses OpenMP for compression/decompression) and the fixest package (which also uses OpenMP for demeaning).

The solution, implemented in `01_inference_fix.R` via `callr::r()`, is to serialize each estimation section as a subprocess: each section launches a fresh R process, completes, saves its RDS files, and exits — releasing all memory and OpenMP handles before the next section starts. This is slower than running sections in parallel but is the only stable approach on Windows with this data configuration.

Rule: Never run more than one heavy R job (anything loading the FST file with fixest models) at the same time on this machine.

10.4 Thread Configuration

Within a single estimation subprocess, the threading configuration is:

```
threads_fst(1) # fst: single-thread (avoids OpenMP conflict)
setFixest_nthreads(detectCores() - 1) # fixest: all cores except 1
```

This means FST reads are serialized (fast anyway since we load only needed columns) while fixest's alternating projections algorithm (demeaning for HDFE) uses all available CPU cores. On the 24-core machine, this gives ~23 threads to fixest, which substantially speeds up the most expensive step in estimation.

11. References

- Abman, R., Lundberg, C., & Ruta, M. (2024).** The effectiveness of environmental provisions in regional trade agreements. *Journal of the European Economic Association*, 22(6), 2507–2548. DOI: 10.1093/jeea/jvae023
- Baccini, L. (2017).** The economics and politics of preferential trade agreements. *Annual Review of Political Science*, 22, 75–92.
- Bertrand, M., Duflo, E., & Mullainathan, S. (2004).** How much should we trust differences-in-differences estimates? *Quarterly Journal of Economics*, 119(1), 249–275.
- Brandi, C., Schwab, J., Berger, A., & Morin, J.-F. (2020).** Do environmental provisions in trade agreements make exports greener? *World Development*, 129, 104899.
- Callaway, B., & Sant'Anna, P. H. C. (2021).** Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2), 200–230.
- Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008).** Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414–427.
- Cherniwchan, J. (2017).** Trade liberalization and the environment: Evidence from NAFTA and U.S. manufacturing. *Journal of International Economics*, 105, 130–149. DOI: 10.1016/j.jinteco.2017.01.005
- Copeland, B. R., Shapiro, J. S., & Taylor, M. S. (2022).** Globalization and the environment. In G. Gopinath, E. Helpman, & K. Rogoff (Eds.), *Handbook of International Economics*, Vol. 5. Elsevier. DOI: 10.1016/bs.hesint.2022.02.002
- de Chaisemartin, C., & D'Haultfoeuille, X. (2020).** Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review*, 110(9), 2964–2996.
- Dechezleprêtre, A., & Sato, M. (2017).** The impacts of environmental regulations on competitiveness. *Review of Environmental Economics and Policy*, 11(2), 183–206.
- Goodman-Bacon, A. (2021).** Difference-in-differences with variation in treatment timing. *Journal of Econometrics*, 225(2), 254–277.
- Larch, M., Shikher, S., & Yotov, Y. V. (2025).** Fifteen recommendations for applied gravity research. *Review of International Economics*, forthcoming.
- Low, P., & Yeats, A. (1992).** Do 'dirty' industries migrate? In P. Low (Ed.), *International Trade and the Environment*. World Bank.
- Mani, M., & Wheeler, D. (1998).** In search of pollution havens? Dirty industry in the world economy, 1960–1995. *Journal of Environment and Development*, 7(3), 215–247.
- Morin, J.-F., Dür, A., & Lechner, L. (2018).** Mapping the trade and environment nexus: Insights from a new dataset. *Global Environmental Politics*, 18(1), 122–139.
- Neri-Laine, A., Orefice, G., & Ruta, M. (2023).** Deep trade agreements and heterogeneous firms' exports. CESifo Working Paper No. 10436.
- Santos Silva, J. M. C., & Tenreyro, S. (2006).** The log of gravity. *Review of Economics and Statistics*, 88(4), 641–658.
- Shapiro, J. S. (2021).** The environmental bias of trade policy. *Quarterly Journal of Economics*, 136(2), 831–886. DOI: 10.1093/qje/qjaa042
- Sun, L., & Abraham, S. (2021).** Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, 225(2), 175–199.

